# Maximizing Early Brain Development in Underserved Children:

## A Feasibility Study of a Caregiver-Centered, Neuroscience-Informed Intervention (Pilot Study)

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## INTRODUCTION

Children from low-socioeconomic status backgrounds face persistent disparities in early language exposure, school readiness, and academic achievement. Early language environments are strongly linked to neural development during sensitive periods, particularly in systems supporting language, executive function, and learning.

Caregiver engagement represents a modifiable pathway through which early developmental disparities may be addressed. Recent advances in developmental neuroscience highlight the importance of early, frequent, and meaningful language interactions for strengthening neural circuits involved in learning. However, scalable, low-burden approaches that engage caregivers in underserved communities remain limited.

This approach is informed by developmental neuroscience principles, including sensitive periods of brain development, experience-dependent plasticity, and the importance of frequent, meaningful language input for strengthening neural circuits supporting learning.

## RESEARCH QUESTION

How feasible and acceptable is a brief, caregiver-centered survey designed to inform the development of a neuroscience-informed early learning intervention for underserved families?

## STUDY OBJECTIVES

1. Evaluate the feasibility of survey administration in an underserved caregiver sample.
2. Assess caregiver perceptions of clarity, burden, and engagement.
3. Examine caregiver interest in participating in a future early learning intervention.
4. Explore the relationship between early learning values and intervention interest.

## METHODS

### Participants

Caregivers of young children were recruited for a pilot survey focused on early learning beliefs, experiences, and intervention interest (N = 28).

Participants primarily represented underserved family contexts.

### Procedure

Participants completed a brief online survey assessing perceptions of early learning, clarity of survey items, perceived burden, and interest in future intervention participation.

Open-ended questions captured qualitative feedback regarding the survey experience.

### Measures

Survey clarity (Likert scale)

Perceived burden (Likert scale)

Interest in future intervention (Likert scale)

Perceived importance of early learning

Open-ended caregiver feedback

### Analytic Strategy

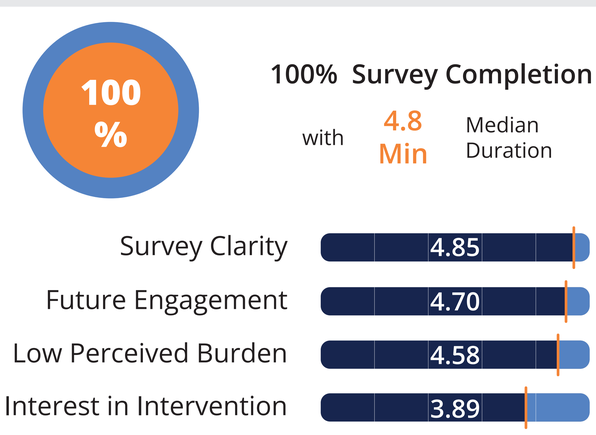
Descriptive statistics were used to assess feasibility and acceptability. An exploratory linear regression examined the association between perceived importance of early learning and intervention interest. Qualitative responses were reviewed for recurring themes.

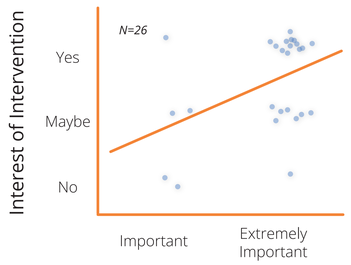
## RESULTS

**100%**

**100% Survey Completion**

with **4.8 Min** Median Duration




**Perceived Importance of Early Learning**

In a regression model ($R^2$ = .50), parent confidence (β = −.34) and perceived barriers (β = .56) were the strongest predictors of intervention interest.

**Caregiver Feedback**

*It was great-more questionnaires should be like this.*

*Very self-explanatory*

*Not that I can think of!*

## DISCUSSION

Findings from this pilot study indicate that a brief, caregiver-centered survey is both feasible and highly acceptable within an underserved sample. High completion rates, short completion times, and strong clarity ratings suggest that the format minimizes burden while maintaining engagement.

This exploratory, hypothesis-generating association between early learning values and intervention interest suggests that caregiver priorities may be meaningfully related to engagement with early learning supports. Although exploratory, these results suggest that caregiver beliefs may play a meaningful role in predicting engagement with early learning supports.

## IMPLICATIONS FOR INTERVENTION DESIGN

These findings support the development of a neuroscience-informed, caregiver-facing intervention that emphasizes:

- Language-rich interactions during sensitive developmental periods
- Clear, accessible guidance for caregivers
- Low-burden digital delivery to support scalability

Language-rich environments are defined by the quality and frequency of caregiver–child interactions, including verbal engagement, labeling, and responsive communication, rather than reliance on shared reading or print-based activities alone.

This work provides foundational evidence to inform the design of an app-based intervention focused on strengthening early language environments through caregiver engagement.

## LIMITATIONS

This pilot study used a small, self-selected sample and relied on self-report measures. Findings are descriptive and exploratory and should be interpreted accordingly. Future research should examine intervention efficacy using longitudinal and experimental designs.

## FUTURE DIRECTIONS

**Next steps include:**

- Development of a caregiver-centered digital intervention informed by neuroscience principles
- Pilot testing intervention usability and engagement
- Longitudinal evaluation of child language and developmental outcomes

## SELECTED REFERENCES

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## CONCLUSION

Caregiver-centered, neuroscience-informed approaches show promise as scalable strategies for supporting early brain development in underserved children. This pilot study establishes feasibility and acceptability, providing a foundation for future intervention research.